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2 **Supplement to “E-Bayesian and H-Bayesian estimation of a single**
3 **server Markovian queue with balking”**

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6 **ARTICLE HISTORY**

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8 **ABSTRACT**

9 This supplementary material presents the complete numerical results supporting the
10 proposed estimation methods for the trac intensity (ρ) and the average queue length
11 (L_q) in the considered queueing model. The results include classical estimators,
12 namely the Maximum Likelihood Estimator (MLE) and Uniformly Minimum Vari-
13 ance Unbiased Estimator (UMVUE), together with Bayesian, E-Bayesian, and Hier-
14 archical Bayesian (H-Bayesian) estimators under a specied prior using the Squared
15 Error Loss Function (SELF) and Precautionary Loss Function (PLF). Bayesian
16 and H-Bayesian estimation based on Markov Chain Monte Carlo (MCMC) under a
17 Gamma prior are also reported. In addition, predictive probabilities obtained using
18 both analytical expressions and the Sampling Importance Resampling (SIR) method
19 are provided for the Bayesian and H-Bayesian approaches. The supplementary ma-
20 terial further includes extensive simulation results and the analysis of a real-life
21 dataset to demonstrate the practical applicability of the proposed methodologies.

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Table 1.: Mean estimates and MSE of ρ using MLE

n	10	20	50	100	150	200
ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.1	0.10190 (0.01113)	0.10105 (0.00487)	0.09980 (0.00200)	0.09865 (0.00100)	0.09963 (0.00064)	0.09998 (0.00054)
0.2	0.19750 (0.01835)	0.20545 (0.01065)	0.20186 (0.00413)	0.19792 (0.00191)	0.20085 (0.00131)	0.19933 (0.00096)
0.5	0.49560 (0.04636)	0.49870 (0.02519)	0.49964 (0.00970)	0.49840 (0.00462)	0.49936 (0.00337)	0.50104 (0.00248)
0.7	0.69450 (0.07221)	0.70660 (0.03248)	0.69606 (0.01462)	0.70123 (0.00636)	0.69994 (0.00474)	0.70242 (0.00343)
0.9	0.91700 (0.09188)	0.89270 (0.04563)	0.90910 (0.01868)	0.90189 (0.00879)	0.90182 (0.00569)	0.89936 (0.00454)
0.99	0.99960 (0.10303)	0.98430 (0.04802)	0.99060 (0.01962)	0.98848 (0.01047)	0.99328 (0.00681)	0.98918 (0.00471)

Table 2.: Mean estimates and MSE of ρ using UMVUE

n	10	20	50	100	150	200
ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.1	0.10222 (0.01062)	0.09985 (0.00498)	0.09812 (0.00213)	0.09954 (0.00094)	0.09914 (0.00063)	0.10015 (0.00050)
0.2	0.19970 (0.01903)	0.19580 (0.00964)	0.20148 (0.00400)	0.19935 (0.00188)	0.20027 (0.00132)	0.19994 (0.00102)
0.5	0.49720 (0.04638)	0.49680 (0.02498)	0.50520 (0.01060)	0.50262 (0.00480)	0.49940 (0.00321)	0.50026 (0.00256)
0.7	0.70380 (0.07336)	0.70330 (0.03541)	0.69284 (0.01466)	0.69819 (0.00683)	0.69810 (0.00479)	0.69875 (0.00364)
0.9	0.91970 (0.09507)	0.89195 (0.04294)	0.90296 (0.01809)	0.90092 (0.009194)	0.89506 (0.00606)	0.89921 (0.00428)
0.99	0.98420 (0.10028)	0.98155 (0.05019)	0.99470 (0.01924)	0.98874 (0.00990)	0.99002 (0.00630)	0.98830 (0.00463)

Table 3.: Mean estimates and MSE of ρ for Bayesian estimation under gamma prior ($\alpha = 2, \beta = 3$)

n		10	20	50	100	150	200
Loss function	ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.1	0.22845 (0.01723)	0.17234 (0.00749)	0.13058 (0.00246)	0.11718 (0.00112)	0.11064 (0.00072)	0.10832 (0.00053)
	0.2	0.30396 (0.02195)	0.25979 (0.01125)	0.22569 (0.00425)	0.21143 (0.00205)	0.20917 (0.00137)	0.20786 (0.00102)
	0.5	0.51275 (0.02928)	0.51087 (0.01964)	0.50632 (0.00944)	0.50572 (0.00491)	0.50237 (0.00328)	0.50213 (0.00247)
	0.7	0.61254 (0.02834)	0.65852 (0.01973)	0.68731 (0.01106)	0.70038 (0.00650)	0.69790 (0.00450)	0.69815 (0.00343)
	0.9	0.70885 (0.02436)	0.76629 (0.01619)	0.82922 (0.00855)	0.86096 (0.00513)	0.87556 (0.00382)	0.88068 (0.00307)
	0.99	0.73517 (0.02267)	0.79859 (0.01441)	0.86994 (0.00673)	0.90591 (0.00368)	0.92254 (0.00255)	0.93313 (0.00193)
PLF	0.1	0.26344 (0.06997)	0.19274 (0.04079)	0.13967 (0.01817)	0.12193 (0.00950)	0.11386 (0.00644)	0.11075 (0.00487)
	0.2	0.33854 (0.06918)	0.28055 (0.04152)	0.23493 (0.01846)	0.21622 (0.00960)	0.21241 (0.00648)	0.21031 (0.00490)
	0.5	0.54191 (0.05831)	0.53021 (0.03867)	0.51558 (0.01853)	0.51054 (0.00966)	0.50563 (0.00651)	0.50458 (0.00491)
	0.7	0.63670 (0.04830)	0.67404 (0.03105)	0.69548 (0.01634)	0.70503 (0.00930)	0.70113 (0.00645)	0.70060 (0.00490)
	0.9	0.72699 (0.03629)	0.77733 (0.02209)	0.83453 (0.01062)	0.86400 (0.00609)	0.87777 (0.00443)	0.88246 (0.00355)
	0.99	0.75147 (0.03260)	0.80801 (0.01884)	0.87394 (0.00798)	0.90798 (0.00416)	0.92395 (0.00282)	0.93419 (0.00210)

Table 4.: Mean estimates and MSE of ρ for E-Bayesian estimation with $c = 1.5$

n		10	20	50	100	150	200
Loss function	ρ	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)
SELF	0.1	* (*)	* (*)	* (*)	0.10408 (0.00103)	0.10294 (0.00068)	0.10292 (0.00051)
	0.2	* (*)	* (*)	0.20788 (0.00410)	0.20005 (0.00198)	0.20228 (0.00134)	0.20198 (0.00101)
	0.5	* (*)	0.50014 (0.02058)	0.50387 (0.00980)	0.50231 (0.004984)	0.49936 (0.00331)	0.49939 (0.00249)
	0.7	0.59876 (0.03038)	0.64917 (0.02103)	0.68646 (0.01146)	0.69902 (0.00658)	0.69886 (0.00458)	0.70264 (0.00348)
	0.9	0.70452 (0.02586)	0.76674 (0.01655)	0.82852 (0.00872)	0.85759 (0.00528)	0.87557 (0.00382)	0.88065 (0.00311)
	0.99	0.73170 (0.02403)	0.79989 (0.01473)	0.86957 (0.00688)	0.90716 (0.00366)	0.92151 (0.00260)	0.93196 (0.00198)
PLF	0.1	* (*)	* (*)	* (*)	0.10892 (0.00968)	0.10620 (0.00652)	0.10538 (0.00492)
	0.2	* (*)	* (*)	0.21748 (0.01921)	0.20495 (0.00979)	0.20557 (0.00657)	0.20446 (0.00495)
	0.5	* (*)	0.52101 (0.04173)	0.51353 (0.01931)	0.50725 (0.00987)	0.50266 (0.00661)	0.50187 (0.00497)
	0.7	0.62584 (0.05415)	0.66602 (0.03369)	0.69495 (0.01697)	0.70375 (0.00945)	0.70214 (0.00654)	0.70311 (0.00495)
	0.9	0.72424 (0.03944)	0.77815 (0.02283)	0.83395 (0.01087)	0.86073 (0.00629)	0.87778 (0.00444)	0.88244 (0.00357)
	0.99	0.74935 (0.03530)	0.80958 (0.01938)	0.87366 (0.00817)	0.90923 (0.00413)	0.92294 (0.00288)	0.93304 (0.00216)

* These values are not available

Table 5.: Mean estimates and MSE of ρ for H-Bayesian estimation with $c = 1.5$

n		10	20	50	100	150	200
Loss function	ρ	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.1	0.12408 (0.01148)	0.11816 (0.00579)	0.10929 (0.00217)	0.10374 (0.00103)	0.10295 (0.00068)	0.10359 (0.00052)
	0.2	0.22573 (0.01971)	0.21302 (0.01032)	0.20599 (0.00407)	0.20206 (0.00201)	0.20227 (0.00134)	0.20254 (0.00101)
	0.5	0.47497 (0.03133)	0.49682 (0.02064)	0.50681 (0.00988)	0.50213 (0.00499)	0.50390 (0.00334)	0.50005 (0.00249)
	0.7	0.60907 (0.03051)	0.65007 (0.02078)	0.68776 (0.01137)	0.69775 (0.00661)	0.69672 (0.00456)	0.70244 (0.00348)
	0.9	0.70723 (0.02588)	0.76772 (0.01666)	0.82867 (0.00868)	0.86194 (0.00518)	0.87269 (0.00384)	0.88419 (0.00309)
	0.99	0.74070 (0.02394)	0.81111 (0.01402)	0.87342 (0.00663)	0.90709 (0.00367)	0.92407 (0.00251)	0.93315 (0.00194)
PLF	0.1	0.15132 (0.05449)	0.13766 (0.03900)	0.11870 (0.018823)	0.10860 (0.00972)	0.10622 (0.00654)	0.10606 (0.00493)
	0.2	0.26135 (0.07126)	0.23536 (0.04470)	0.21563 (0.01929)	0.20697 (0.00982)	0.20556 (0.00659)	0.20502 (0.00496)
	0.5	0.50881 (0.06768)	0.51788 (0.04212)	0.51650 (0.01936)	0.50707 (0.00988)	0.50721 (0.00662)	0.50254 (0.00497)
	0.7	0.63573 (0.05332)	0.66680 (0.03346)	0.69619 (0.01685)	0.70251 (0.00951)	0.69999 (0.00655)	0.70492 (0.00497)
	0.9	0.72689 (0.03932)	0.77916 (0.02287)	0.83409 (0.01083)	0.86501 (0.00614)	0.87493 (0.00448)	0.88596 (0.00351)
	0.99	0.75787 (0.03434)	0.82018 (0.01815)	0.87736 (0.00786)	0.90916 (0.00414)	0.92545 (0.00276)	0.93421 (0.00211)

Table 6.: Mean estimates and MSE of L_q using MLE

n	10	20	50	100	150	200
L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.15	0.16854 (0.01341)	0.15926 (0.00671)	0.15469 (0.00244)	0.15287 (0.00128)	0.15026 (0.00083)	0.15036 (0.00060)
0.20	0.21190 (0.01888)	0.20519 (0.00868)	0.20470 (0.00358)	0.20092 (0.00179)	0.20241 (0.00117)	0.20221 (0.00096)
0.25	0.27117 (0.02656)	0.26065 (0.01271)	0.25163 (0.00464)	0.25127 (0.00247)	0.250819 (0.00163)	0.25057 (0.00126)
0.30	0.32079 (0.03176)	0.30376 (0.01637)	0.30416 (0.00585)	0.30308 (0.00330)	0.29970 (0.00182)	0.30071 (0.00165)
0.35	0.37739 (0.03911)	0.35926 (0.01770)	0.35122 (0.00781)	0.35287 (0.00377)	0.35328 (0.00248)	0.35211 (0.00180)

Table 7.: Mean estimates and MSE of L_q using UMVUE

n	10	20	50	100	150	200
L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
0.15	0.15354 (0.01268)	0.14840 (0.00605)	0.15065 (0.00246)	0.15059 (0.00123)	0.14949 (0.00083)	0.15003 (0.00062)
0.20	0.20152 (0.01893)	0.19551 (0.00824)	0.19853 (0.00327)	0.19971 (0.00192)	0.19940 (0.00117)	0.20075 (0.00085)
0.25	0.25594 (0.02470)	0.25110 (0.01175)	0.25507 (0.00574)	0.25046 (0.00254)	0.24990 (0.00159)	0.25038 (0.00127)
0.30	0.29784 (0.03123)	0.29949 (0.01454)	0.30212 (0.00675)	0.30023 (0.00318)	0.29936 (0.00182)	0.30079 (0.00157)
0.35	0.34379 (0.03466)	0.35909 (0.02009)	0.35307 (0.00798)	0.34754 (0.00383)	0.35247 (0.00258)	0.35008 (0.00202)

Table 8.: Mean estimates and MSE of L_q for Bayesian estimation under gamma prior ($\alpha = 2, \beta = 3$)

n		10	20	50	100	150	200
Loss function	L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.15	0.15012 (0.00562)	0.15274 (0.00415)	0.15650 (0.00232)	0.15265 (0.00124)	0.15156 (0.00082)	0.15192 (0.00062)
	0.20	0.16943 (0.00585)	0.18549 (0.00449)	0.20036 (0.00280)	0.20023 (0.00168)	0.20031 (0.00118)	0.20176 (0.00090)
	0.25	0.18674 (0.00585)	0.21008 (0.00444)	0.23537 (0.00286)	0.24592 (0.00184)	0.24757 (0.00139)	0.24836 (0.00111)
	0.30	0.20790 (0.00573)	0.23232 (0.00433)	0.26303 (0.00262)	0.27930 (0.00171)	0.28662 (0.00130)	0.29063 (0.00106)
	0.35	0.21969 (0.00557)	0.25158 (0.00403)	0.28757 (0.00221)	0.30684 (0.00132)	0.31542 (0.00097)	0.32136 (0.00077)
PLF	0.15	0.16863 (0.03702)	0.16609 (0.02668)	0.16373 (0.01445)	0.15659 (0.00787)	0.15424 (0.00534)	0.15394 (0.00404)
	0.20	0.18690 (0.03494)	0.19783 (0.02467)	0.20737 (0.01401)	0.20439 (0.00831)	0.20321 (0.00581)	0.20398 (0.00442)
	0.25	0.20291 (0.03234)	0.22110 (0.02204)	0.24161 (0.01248)	0.24973 (0.00763)	0.25040 (0.00565)	0.25059 (0.00447)
	0.30	0.22234 (0.02888)	0.24209 (0.01955)	0.26822 (0.01039)	0.28245 (0.00630)	0.28894 (0.00464)	0.29249 (0.00372)
	0.35	0.23306 (0.02674)	0.25999 (0.01682)	0.29158 (0.00803)	0.30907 (0.00446)	0.31702 (0.00318)	0.32260 (0.00247)

Table 9.: Mean estimates and MSE of L_q for E-Bayesian estimation with $c = 1.5$

n		10	20	50	100	150	200
Loss function	L_q	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)	Mean (E-MSE)
SELF	0.15	* (*)	0.15168 (0.00431)	0.15462 (0.00237)	0.15369 (0.00128)	0.15250 (0.00085)	0.15225 (0.00063)
	0.20	* (*)	0.18447 (0.00467)	0.19707 (0.00286)	0.20082 (0.00171)	0.20237 (0.00120)	0.19985 (0.00090)
	0.25	0.18572 (0.00615)	0.20950 (0.00469)	0.23735 (0.00291)	0.24716 (0.00187)	0.24914 (0.00140)	0.24868 (0.00112)
	0.30	0.20777 (0.00595)	0.23328 (0.00449)	0.26363 (0.00264)	0.28154 (0.00170)	0.28633 (0.00130)	0.29124 (0.00107)
	0.35	0.21835 (0.00583)	0.25190 (0.00409)	0.28692 (0.00223)	0.30860 (0.00130)	0.31634 (0.00096)	0.32256 (0.00076)
PLF	0.15	* (*)	0.16575 (0.02854)	0.16206 (0.01489)	0.15773 (0.00808)	0.15522 (0.00543)	0.15429 (0.00409)
	0.20	* (*)	0.19743 (0.02592)	0.20434 (0.01455)	0.20504 (0.00844)	0.20531 (0.00531)	0.20209 (0.00447)
	0.25	0.20301 (0.03459)	0.22118 (0.02336)	0.24365 (0.01262)	0.25101 (0.00771)	0.25198 (0.00568)	0.25094 (0.00452)
	0.30	0.22296 (0.03039)	0.24342 (0.02027)	0.26887 (0.01048)	0.28466 (0.00624)	0.28866 (0.00465)	0.29311 (0.00374)
	0.35	0.23257 (0.02845)	0.26053 (0.01726)	0.29100 (0.00812)	0.31078 (0.00436)	0.31790 (0.00312)	0.32374 (0.00243)

* These values are not available

Table 10.: Mean estimates and MSE of L_q for H-Bayesian estimation for $c = 1.5$

n		10	20	50	100	150	200
Loss function	L_q	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)	Mean (MSE)
SELF	0.15	* (*)	0.15223 (0.00440)	0.15780 (0.00243)	0.15193 (0.00125)	0.15226 (0.00084)	0.15171 (0.00063)
	0.20	0.17213 (0.00607)	0.18414 (0.00470)	0.19725 (0.00283)	0.20692 (0.00172)	0.20120 (0.00120)	0.20287 (0.00092)
	0.25	0.18805 (0.00604)	0.21685 (0.00462)	0.23634 (0.00288)	0.24798 (0.00188)	0.24858 (0.00140)	0.25023 (0.00112)
	0.30	0.20874 (0.00600)	0.23663 (0.00445)	0.26669 (0.00261)	0.28249 (0.00170)	0.28869 (0.00129)	0.29230 (0.00106)
	0.35	0.22382 (0.00573)	0.25281 (0.00408)	0.28688 (0.00222)	0.30699 (0.00013)	0.31655 (0.00097)	0.32237 (0.00077)
PLF	0.15	* (*)	0.16648 (0.02851)	0.16546 (0.01493)	0.15594 (0.00803)	0.15498 (0.00543)	0.15375 (0.00408)
	0.20	0.19047 (0.03668)	0.19721 (0.02613)	0.20446 (0.01442)	0.20692 (0.00845)	0.20414 (0.00588)	0.20511 (0.00448)
	0.25	0.20506 (0.03402)	0.22807 (0.02243)	0.24263 (0.01259)	0.25183 (0.00770)	0.25143 (0.00569)	0.25248 (0.00450)
	0.30	0.22399 (0.03051)	0.24655 (0.01984)	0.27182 (0.01026)	0.28560 (0.00621)	0.29099 (0.00460)	0.29416 (0.00371)
	0.35	0.23752 (0.02740)	0.26138 (0.01714)	0.29096 (0.00816)	0.30924 (0.00450)	0.31812 (0.00314)	0.32359 (0.00243)

* These values are not available

Table 11.: Bayesian Estimation for ρ under gamma prior

ρ	n	Mean (MSE)	Acc-rate	ESS	Geweke-Z score	Convergence
0.3	100	0.32030 (0.00342)	0.90669	275.579	1.25151	convergent
	200	0.28620 (0.00148)	0.88039	450.013	0.56360	convergent
	500	0.29411 (0.00064)	0.81399	937.766	0.67978	convergent
0.5	100	0.50432 (0.00481)	0.88274	399.346	0.73252	convergent
	200	0.52688 (0.00313)	0.84184	744.796	1.08707	convergent
	500	0.48756 (0.00116)	0.76443	1206.006	0.30743	convergent
0.7	100	0.65974 (0.00795)	0.86884	506.251	1.11855	convergent
	200	0.68492 (0.00341)	0.81844	890.420	0.57788	convergent
	500	0.68545 (0.00167)	0.72657	1411.824	0.20919	convergent

Table 12.: H-Estimation for ρ under gamma prior

ρ	n	Mean (MSE)	Acc-rate	ESS	Geweke-Z score	Convergence
0.3	100	0.31264 (0.00323)	0.90800	259.305	1.21991	convergent
	200	0.28191 (0.00158)	0.88294	451.090	0.44234	convergent
	500	0.29411 (0.00064)	0.81399	937.766	0.67978	convergent
0.5	100	0.50158 (0.00493)	0.88589	389.404	0.72554	convergent
	200	0.52613 (0.00311)	0.84239	899.376	0.69573	convergent
	500	0.48691 (0.00118)	0.76658	1180.406	0.24362	convergent
0.7	100	0.65956 (0.00806)	0.86859	502.611	1.19893	convergent
	200	0.68609 (0.00339)	0.81874	899.398	0.77673	convergent
	500	0.68551 (0.00169)	0.72793	1403.149	0.27330	convergent

Table 13.: Comparision between different estimation method along with different loss function

Loss function	ρ	Method	Mean (MSE)	Bias	95% CI
SELF	0.1	Bayesian	0.11231 (0.00055)	0.01231	0.11096-0.11366
		E-Bayesian	0.10104 (0.00048)	0.00104	0.09967-0.10241
		H-Bayesian	0.10311 (0.00052)	0.00311	0.10170-0.10451
PLF	0.1	Bayesian	0.11474 (0.00048)	0.01473	0.11339-0.11608
		E-Bayesian	0.10350 (0.00050)	0.00350	0.10213-0.10487
		H-Bayesian	0.10557 (0.00054)	0.00557	0.10413-0.10697
SELF	0.5	Bayesian	0.50468 (0.00247)	0.00468	0.50164-0.50772
		E-Bayesian	0.49616 (0.00236)	-0.00378	0.49321-0.49921
		H-Bayesian	0.50203 (0.00242)	0.00203	0.49898-0.50508
PLF	0.5	Bayesian	0.50713 (0.00489)	0.00713	0.50409-0.51017
		E-Bayesian	0.49870 (0.00234)	-0.00129	0.49569-0.50170
		H-Bayesian	0.50452 (0.00244)	0.00452	0.50147-0.50757
SELF	0.9	Bayesian	0.88112 (0.00310)	-0.01887	0.87822-0.88401
		E-Bayesian	0.88221 (0.00262)	-0.01778	0.87917-0.88526
		H-Bayesian	0.88359 (0.00246)	-0.01640	0.88069-0.88650
PLF	0.9	Bayesian	0.88290 (0.00356)	-0.01709	0.88003-0.88576
		E-Bayesian	0.88398 (0.00261)	-0.01601	0.88097-0.88703
		H-Bayesian	0.88536 (0.00236)	-0.01463	0.88249-0.88824

Table 14.: Posterior predictive probabilities under Bayesian and H-Bayesian estimation

ρ	m	Bayesian	H-Bayesian	ρ	m	Bayesian	H-Bayesian
0.6	0	0.53365	0.52332	0.8	0	0.48935	0.48768
	1	0.33353	0.33721		1	0.34806	0.34590
	2	0.10583	0.11030		2	0.12544	0.12800
	3	0.02272	0.02441		3	0.03053	0.03166
	4	0.00371	0.00411		4	0.00565	0.00595
	5	0.00049	0.00056		5	0.00084	0.00090
	6	0.00006	0.00006		6	0.00011	0.00012
	≥ 7	0.00000	0.00000		7	0.00001	0.00001
					≥ 8	0.00000	0.00000

Table 15.: Posterior predictive probabilities under Bayesian estimation using SIR method

ρ	m	$\mathcal{B}(1, 2)$	$\mathcal{B}(2, 5)$	$\mathcal{B}(5, 2)$	$\mathcal{B}(7, 3)$
0.6	0	0.54896	0.55141	0.55149	0.55163
	1	0.32612	0.32521	0.32511	0.32494
	2	0.09994	0.09890	0.09889	0.09888
	3	0.02103	0.02064	0.02066	0.02069
	4	0.00341	0.00332	0.00333	0.00334
	5	0.00045	0.00044	0.00044	0.00044
	6	0.00005	0.00005	0.00005	0.00005
	≥ 7	0.00000	0.00000	0.00000	0.00000
0.8	0	0.45725	0.46349	0.45947	0.45946
	1	0.35505	0.35429	0.35453	0.35449
	2	0.14062	0.13755	0.13958	0.13959
	3	0.03782	0.03611	0.03732	0.03735
	4	0.00775	0.00720	0.00762	0.00762
	5	0.00129	0.00116	0.00126	0.00126
	6	0.00018	0.00015	0.00016	0.00018
	≥ 8	0.00000	0.00000	0.00000	0.00000

Table 16.: Posterior predictive probabilities under H-Bayesian estimation using SIR method

ρ	m	$\mathcal{B}(1, 2)$	$\mathcal{B}(2, 5)$	$\mathcal{B}(5, 2)$	$\mathcal{B}(7, 3)$
0.6	0	0.54946	0.55194	0.55211	0.55220
	1	0.32581	0.32489	0.32475	0.32461
	2	0.09977	0.09873	0.09867	0.09868
	3	0.02100	0.02062	0.02062	0.02065
	4	0.00341	0.00332	0.00332	0.00334
	5	0.00046	0.00044	0.00044	0.00044
	6	0.00005	0.00005	0.00005	0.00005
	≥ 7	0.00000	0.00000	0.00000	0.00000
0.8	0	0.45616	0.46317	0.45842	0.45841
	1	0.35523	0.35436	0.35472	0.35468
	2	0.14115	0.13770	0.14008	0.14010
	3	0.03809	0.03618	0.03758	0.03761
	4	0.00783	0.00722	0.00769	0.00770
	5	0.00131	0.00117	0.00128	0.00128
	6	0.00018	0.00016	0.00018	0.00018
	≥ 8	0.00000	0.00000	0.00000	0.00000

Table 17.: Comparison between theoretical and empirical stationary probabilities

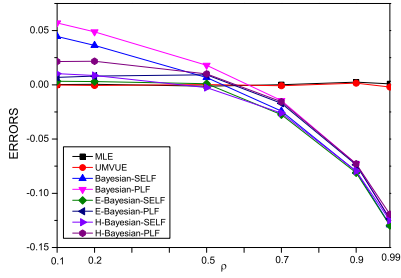
State	Empirical	Theoretical
0	0.54000	0.53830
1	0.31000	0.30430
2	0.07000	0.07345
3	0.07000	0.07345
4	0.01000	0.01049

Table 18.: Mean estimate and squared difference (SD) of ρ and L_q for different estimation methods

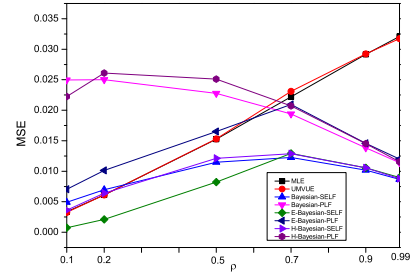
		$\rho_{obs} = 0.73886$	$L_{q_{obs}} = 0.21652$
Methods	Loss function	$\hat{\rho}$ (SD)	$\hat{L}_q$ (SD)
MLE		0.73844 (0.00738)	0.21862 (0.00202)
UMVUE		0.73844 (0.00738)	0.21682 (0.00204)
Bayesian	SELF	0.73114 (0.00650)	0.21388 (0.00176)
	PLF	0.73563 (0.00896)	0.21800 (0.00823)
E-Bayesian	SELF	0.73762 (0.00667)	0.21719 (0.00182)
	PLF	0.74218 (0.00911)	0.22138 (0.00838)
H-Bayesian	SELF	0.73976 (0.00665)	0.21801 (0.00182)
	PLF	0.74430 (0.00907)	0.22219 (0.00836)

Table 19.: Estimation of ρ using the MCMC method

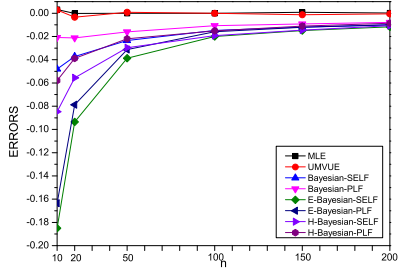
ρ_{obs}		Mean (SD)	Acc-rate	ESS	Geweke-Z-Score	Convergence
0.73886	Bayesian	0.67902 (0.01008)	0.86664	425.2735	1.40842	convergent
0.73886	H-Bayesian	0.67178 (0.01126)	0.81214	924.8394	0.33621	convergent



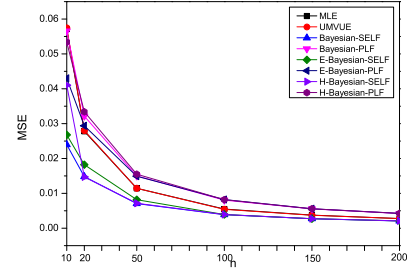
(a) Average errors vs ρ



(b) Average MSEs vs ρ

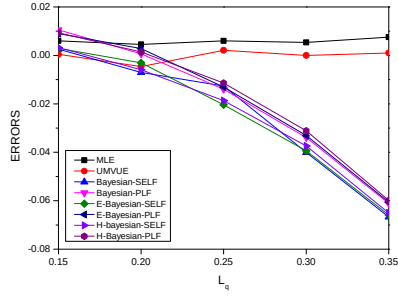


(c) Average errors vs n

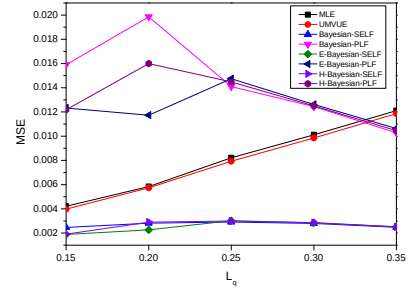


(d) Averages MSEs vs n

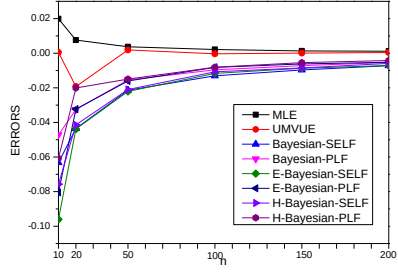
Figure 1.: Performances of the estimators for ρ



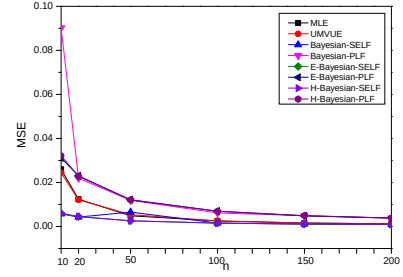
(a) Average errors vs L_q



(b) Average MSEs vs L_q

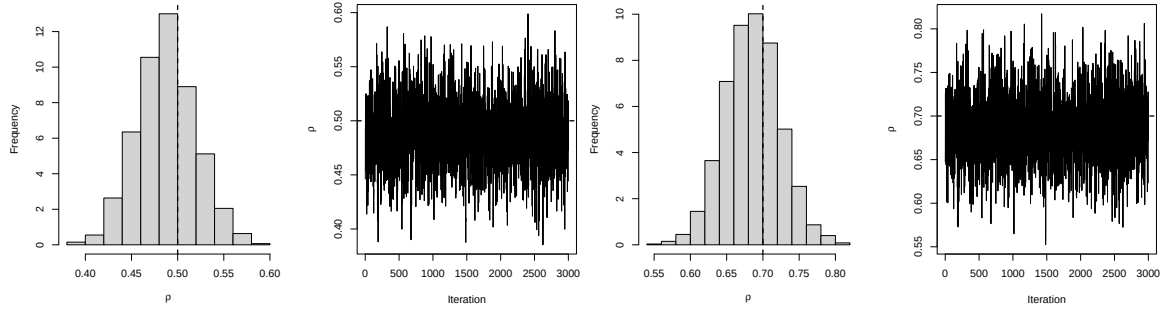


(c) Average errors vs n



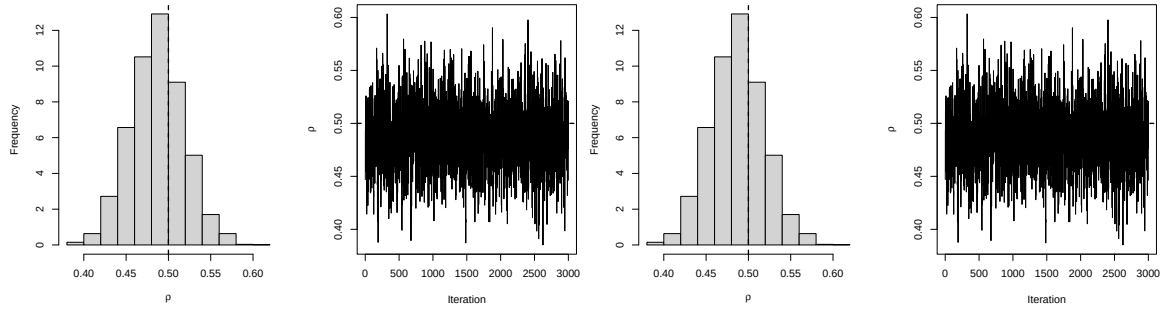
(d) Averages MSEs vs n

Figure 2.: Performances of the estimators of L_q



(a) Trace plot and histogram for $\rho = 0.5$ under Bayesian estimation

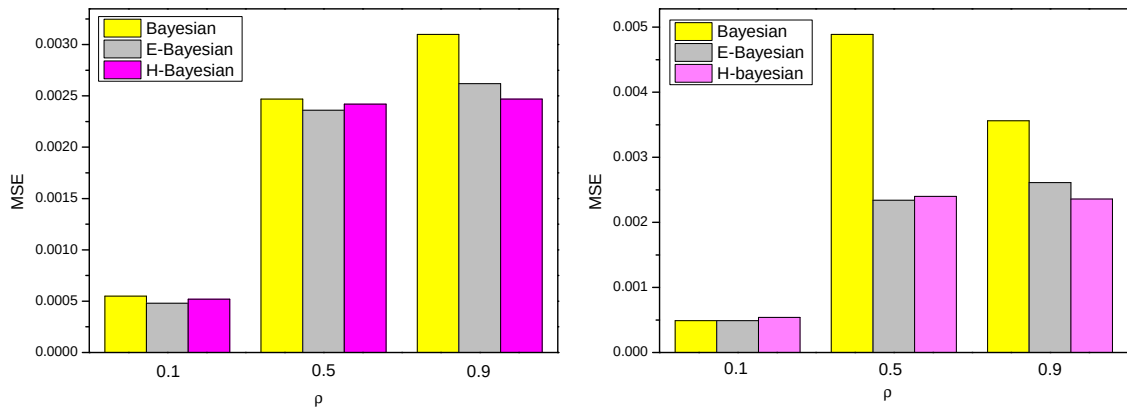
(b) Trace plot and histogram for $\rho = 0.7$ under Bayesian estimation



(c) Trace plot and histogram for $\rho = 0.5$ under H-Bayesian estimation

(d) Trace plot and histogram for $\rho = 0.7$ under H-Bayesian estimation

Figure 3.: Trace plot and histogram for ρ under MCMC method



(a) ρ vs MSE for SELF

(b) ρ vs MSE for PLF

Figure 4.: Comparison between ρ and MSE under different loss function

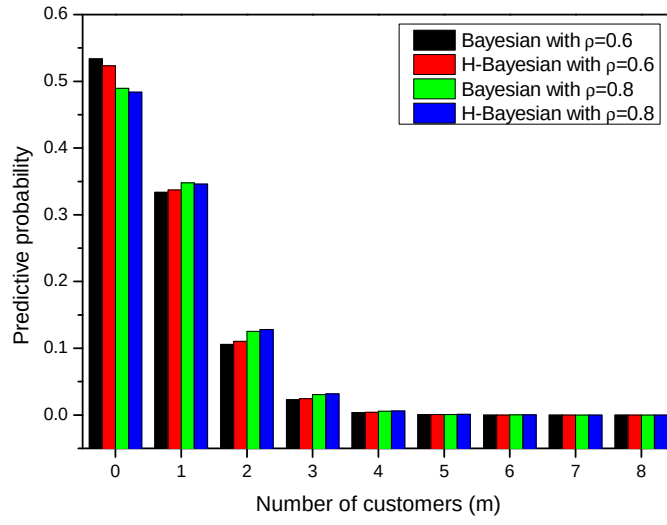


Figure 5.: Posterior predictive probability

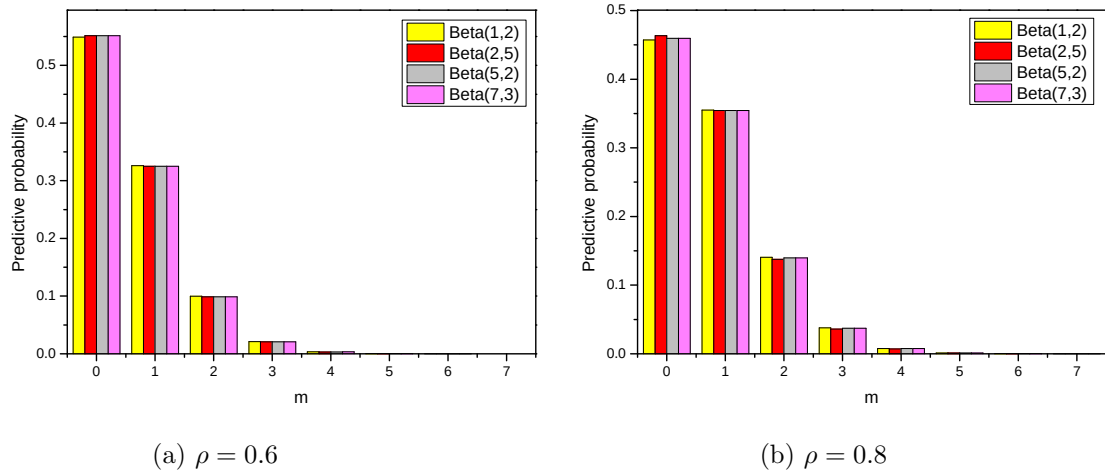


Figure 6.: Posterior predictive distribution for Bayesian estimation using SIR method

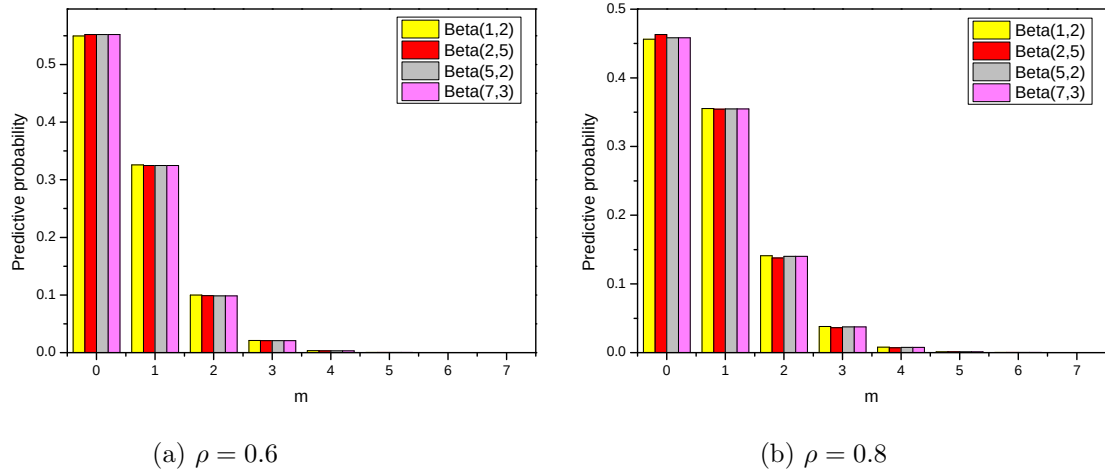


Figure 7.: Posterior predictive distribution for H-Bayesian estimation using SIR method

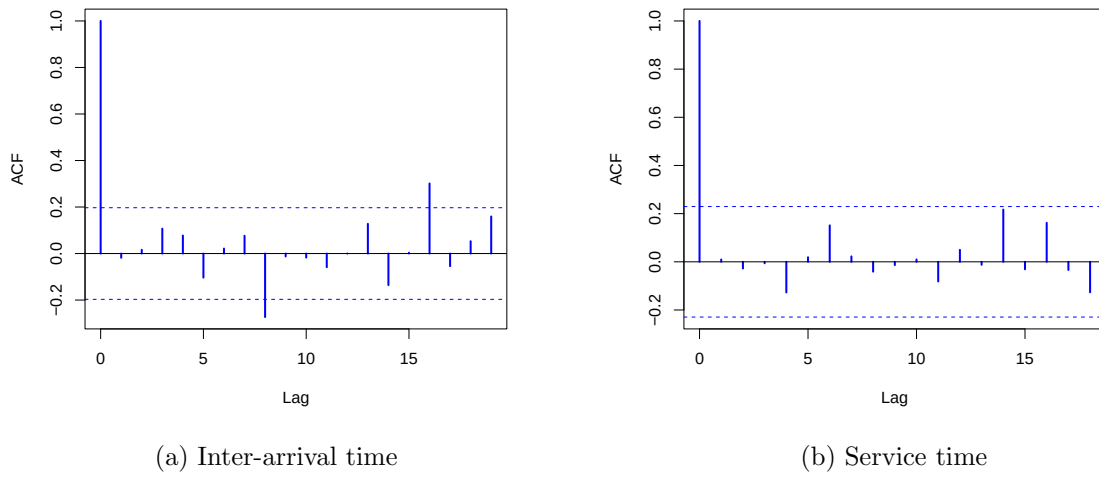


Figure 8.: Autocorrelation function versus lag